



1) Calcula los siguientes límites de varias variables, eliminando las indeterminaciones que se presenten.

$$\lim_{(x,y) \rightarrow (0,0)} \frac{3x^2y}{x^2 + y^2}$$

COORDENADAS POLARES
 $(x = r \cos \theta, y = r \sin \theta)$, DONDE SI $(x,y) \rightarrow (0,0)$, $r \rightarrow 0$:

$$\lim_{r \rightarrow 0} \frac{3(r \cos \theta)^2 (r \sin \theta)}{(r \cos \theta)^2 + (r \sin \theta)^2}$$

$$\lim_{r \rightarrow 0} \frac{3r^3 \cos^2 \theta \sin \theta}{r^2 \cos^2 \theta + r^2 \sin^2 \theta}$$

$$\lim_{r \rightarrow 0} \frac{3r^3 \cos^2 \theta \sin \theta}{r^2(1)} = \lim_{r \rightarrow 0} 3r \cos^2 \theta \sin \theta = 3(0) \cos^2 \theta \sin \theta = 0 //$$

$$\lim_{(x,y) \rightarrow (5,-2)} x^5 + 4x^3y - 5xy^2$$
$$5^5 + 4(5)^3(-2) - 5(5)(-2)^2$$
$$5^5 = 3125$$
$$4 \cdot 125 \cdot (-2) = -1000$$
$$-5 \cdot 5 \cdot 4 = -100$$
$$3125 - 1000 - 100 = 2025 //$$

$$\lim_{(x,y) \rightarrow (6,3)} xy \cos(x - 2y)$$
$$x = 6 \text{ e } y = 3:$$
$$= (6)(3) \cos(6 - 2(3))$$
$$= 18 \cos(6 - 6) = 18 \cos(0)$$
$$\cos(0) = 1$$
$$18(1) = 18 //$$

$$\lim_{\substack{(x,y) \rightarrow (1,1) \\ y \neq x}} \frac{x^2 - y^2}{x - y}$$
$$\frac{(x-y)(x+y)}{x-y} = x + y \rightarrow 1 + 1 = 2 //$$
$$= 2 //$$



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$$\lim_{\substack{(x,y) \rightarrow (2,0) \\ y \neq 2x+4}} \frac{\sqrt{2x-y}-2}{2x-y-4}$$

$$u = \sqrt{2x-y} \rightarrow 2$$

$$\frac{u-2}{u^2-4} = \frac{1}{u+2} \rightarrow \frac{1}{4}$$

$$= \frac{1}{4} //$$

$$\lim_{\substack{(x,y) \rightarrow (0,0) \\ y \neq x}} \frac{x-y+2\sqrt{x}-2\sqrt{y}}{\sqrt{x}-\sqrt{y}}$$

$$\frac{(\sqrt{x}-\sqrt{y})(\sqrt{x}+\sqrt{y}+2)}{\sqrt{x}-\sqrt{y}} =$$

$$\sqrt{x}+\sqrt{y}+2 \rightarrow 2$$

$$= 2 //$$

$$\lim_{(x,y) \rightarrow (0,0)} \frac{8x^2y^2}{x^2+y^4}$$

$$x=0: \frac{0}{y^4} \rightarrow 0$$

$$x=y^2: \frac{8y^4 \cdot y^2}{y^4+y^2} = 4y^2 \rightarrow 0$$

$$0 \leq \frac{8x^2y^2}{x^2+y^4} \leq 8y^2 \rightarrow 0 = 0 //$$

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^3+xy^2}{x^2+y^2}$$

$$\frac{x(x^2+y^2)}{x^2+y^2} = x \rightarrow 0 = 0 //$$



$$\begin{aligned} \lim_{(x,y) \rightarrow (0,0)} \frac{xy}{\sqrt{x^2 + y^2}} \\ &= \lim_{r \rightarrow 0} \frac{(r \cos \theta)(r \sin \theta)}{\sqrt{r^2}} \\ &= \lim_{r \rightarrow 0} \frac{r^2 \cos \theta \sin \theta}{r} = \\ &= \lim_{r \rightarrow 0} r \cos \theta \sin \theta = 0 // \end{aligned}$$

$$\begin{aligned} \lim_{(x,y) \rightarrow (0,0)} \frac{xy + 1}{x^2 + y^2 + 1} \\ \text{DESARROLLAR } x=0 \text{ y } y=0 \\ &= \frac{(0)(0) + 1}{0^2 + 0^2 + 1} \\ &= \frac{0 + 1}{0 + 0 + 1} = \frac{1}{1} = 1 // \end{aligned}$$

$$\begin{aligned} \lim_{\substack{(x,y) \rightarrow (4,3) \\ y \neq x-1}} \frac{\sqrt{x} - \sqrt{y+1}}{x - y - 1} \\ &= \lim_{(x,y) \rightarrow (4,3)} \frac{(\sqrt{x} - \sqrt{y+1})(\sqrt{x} + \sqrt{y+1})}{(x - y - 1)(\sqrt{x} + \sqrt{y+1})} \\ &= \frac{(\sqrt{x})^2 - (\sqrt{y+1})^2}{(x - y - 1)(\sqrt{x} + \sqrt{y+1})} = \frac{x - (y+1)}{(x - y - 1)(\sqrt{x} + \sqrt{y+1})} \\ &= \lim_{(x,y) \rightarrow (4,3)} \frac{x - y - 1}{(x - y - 1)(\sqrt{x} + \sqrt{y+1})} = \lim_{(x,y) \rightarrow (4,3)} \frac{1}{\sqrt{x} + \sqrt{y+1}} \\ &= \frac{1}{\sqrt{4} + \sqrt{3+1}} = \frac{1}{2+2} = \frac{1}{4} // \end{aligned}$$

$$\begin{aligned} \lim_{\substack{(x,y) \rightarrow (2,2) \\ y \neq -x+4}} \frac{x+y-4}{\sqrt{x+y}-2} \\ &= \frac{4-4}{\sqrt{4}-2} = \frac{0}{0} \\ w = x+y &= (x,y) \rightarrow (2,2) \Rightarrow w \rightarrow 2+2=4 \\ \lim_{w \rightarrow 4} \frac{w-4}{\sqrt{w}-2} &= (w-4) = (\sqrt{w}-2)(\sqrt{w}+2) \\ \lim_{w \rightarrow 4} \frac{(\sqrt{w}-2)(\sqrt{w}+2)}{\sqrt{w}-2} &= \lim_{w \rightarrow 4} (\sqrt{w}+2) = w=4 \\ \sqrt{4}+2 &= 2+2=4 // \end{aligned}$$